Model 1 — raw-grain anomaly detection: DESIGN SPECIFICATION

SADINhub Energy Intelligence · DIOM Facility, Riyadh · Model 1 FROZEN 2026-09-20 · 39/39 acceptance, 5/5 reconciliation, 127 tests

Status: signed off 2026-09-20 and IMPLEMENTED. Sections 1-10 are the design as approved; §11 is the implementation record, including four defects found during implementation, one acceptance criterion deliberately relaxed, and a consequence of the severity definition worth revisiting. Every number below was measured read-only against the live database and is reproducible from the queries named in §2.

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0. Corrections to the original Task 1 instructions

The original `TASK 1 — src/model1_anomaly_raw/` instructions cannot be executed as written. Six defects, each verified against the live schema:

#	Original instruction	Problem	Correction
1	`SELECT active_power_kw1, power_factor_1, current_amp1, …`	**All 15 measurement column names are wrong.** None exists.	§2
2	`gold.energy_raw` selected alone	`gold.energy_raw` has **no** `meter_label`; events cannot be labelled without a join	join `gold.meter`
3	`assert count(*) == 4_535_918`	Correct for 31 meters, but wrong if Model 1 adopts Model 2's 29-meter scope	§1 — both asserts defined
4	`KNOWN_FINDINGS` = 6 meters	Incomplete (misses two large findings) and partly unsupported by the data	§3
5	`exclude_windows={"ZN.16-PB": full history}`	Excluding ZN.16's full history leaves **no baseline at all**	§6
6	`cusum_flags(pf, target=0.8)` with no load gate	Would fire on **28.38%** of all phase readings — 97.10% of it low-load noise	§7

Defect 6 is the important one. The instruction to detect "sustained drift below the acceptable threshold" is sound; applied without a load gate it produces an alarm on more than a quarter of the dataset and is therefore not a detector.

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1. Scope, and the row-count assertions

Zone 13's two meters are excluded from *validated consumption* because of cumulative-kWh integrity concerns. That is a **register** concern, not an electrical-measurement concern. Voltage, current, power and power factor for those meters are as trustworthy as any other. Model 1 therefore uses two scopes:

Scope	Meters	Readings	Used by	Verified
`ELECTRICAL` — all canonical meters	31	**4,535,918**	families T1, T2, T3, T4	yes
`REGISTER` — Zone 13 excluded	29	**4,260,849**	family T5	yes

Both counts were confirmed live. Each detector run must assert its own scope's count and fail loudly on a mismatch; the single hard-coded `4_535_918` in the original instruction is correct only for the first.

> **Decision 1, approved 2026-09-20:** the electrical families run on all 31 meters, the register family on 29. Both counts are asserted at the start of every run.

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2. Exact live database columns and joins

2.1 The one join Model 1 needs

```
FROM gold.energy_raw            e
JOIN gold.meter                 m USING (energy_meter_key)
LEFT JOIN gold.zone             z USING (zone_key)
LEFT JOIN silver.dim_meter_cb_rating r USING (energy_meter_key)
```

`silver.dim_meter_cb_rating` is a documented boundary exception: `cb_max_rating_a` is exposed by no Gold view and every current- and PF-related rule depends on it. Read-only, as everywhere else in `ML_WORK`.

2.2 Columns, by the names that actually exist

Purpose	Live column (`gold.energy_raw`)	Type	Original instruction said
Active power, phase *k*	`active_power_phase_1_kw` … `_3_kw`	double	~~ `active_power_kw1` ~~
Active power, total	`active_power_total_kw`	double	~~ `TkW` ~~
Current, phase *k*	`current_phase_1_a` … `_3_a`	double	~~ `current_amp1` ~~
Power factor, phase *k*	`power_factor_phase_1` … `_3`	double	~~ `power_factor_1` ~~
Power factor, average	`power_factor_average`	double	~~ `PFa` ~~
Reactive power	`reactive_power_phase_1_kvar` … `_3_kvar` , `reactive_power_total_kvar`	double	not used
Cumulative active energy	`active_energy_cumulative_kwh`	double	not used
Timestamp	`reading_timestamp_utc`	timestamp	✓
Current status, phase *k*	`status_amp1` , `status_amp2` , `status_amp3`	text	✓
PF status	`status_pf1` , `status_pf2` , `status_pf3` , `status_pfa`	text	✓
Energy status	`status_kwh`	text	not used
Roll-up	`overall_status`	text	not used

Purpose	Live column (`gold.energy_raw`)	Type	Original instruction said
Meter identity	`gold.meter.meter_label` , `.phase_count` , `.maximum_kwh`	—	—
Breaker rating	`silver.dim_meter_cb_rating.cb_max_rating_a`	double	—

Saudi-local day = `reading_timestamp_utc + interval '3 hours'` ; the Saudi day closes at `D 21:00:00 UTC` . No DST.

2.3 Columns that are empty and must not be used

Verified over the **full** history (4,535,918 rows), which the pipeline documentation records as not formally completed:

Column	Populated
`voltage_line_average_v`	**0**
`voltage_neutral_average_v`	**0**
`apparent_power_phase_1_kva` / `_2_` / `_3_`	**0**
`apparent_power_total_kva`	**0**
`apparent_energy_cumulative_kvah`	**0**

Consequence: **no load-weighted facility power factor can be computed from telemetry**, and any detector needing apparent power must derive it as `kW / PF` with an explicit divide-by-small-PF guard. All other measurement columns are 100% populated.

2.4 Two traps in the existing status columns

1. `status_amp1/2/3` **are excluded from** `overall_status` **.** `overall_status` is the worst of exactly 12 inputs — six phase voltages, frequency, four power factors, and accumulated energy. The three current statuses are *deliberately* not among them. `Major Issue` and `Major Problem` must be read from `status_amp*` directly or they are invisible.
2. `overall_status` **is, in practice, a power-factor alarm.** Measured over the 29-meter scope:

`overall_status`	readings	share
Critical	2,623,574	61.57%
Acceptable	814,490	19.12%
Problematic	781,994	18.35%
Review Required	40,766	0.96%
Invalid	25	0.00%

Of the 2,623,574 Critical readings, **100.00% are Critical because of a power-factor status**. Voltage contributes 0, frequency 0, accumulated energy 0. Combined with §7 (97.10% of PF-Critical occurs below 15% of breaker rating), `overall_status` cannot be used as an anomaly label or as a corroborating signal without load gating.

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3. Known-finding acceptance set

The acceptance set is the list of meter/phase findings a working Model 1 **must** detect. It is tiered by evidence strength, because "engineering-confirmed" and "the data shows a pattern" are not the same claim.

Tier A — engineering-confirmed

Meter	Phases	Signature	Evidence	Window
ZN.16-PB	1, 2, 3	Reverse-power / polarity, all three phases	`active_power_phase_k_kw < 0` on **95.37%** of readings on every phase (143,775 / 143,774 / 143,774), `power_factor_phase_k < 0` on the same rows, `current_phase_k_a` positive throughout	2024-12-15 → **2026-06-26**

Evidence source: pipeline documentation §13 ("engineering-confirmed wiring/polarity fault affecting all three phases"), corroborated by direct measurement.

Tier B — strong empirical, not engineering-confirmed

Meter	Phase	% readings with negative kW	n	Window	Documented?
ZN.8-PB.2	1	96.71%	153,413	2024-12-10 → 2026-06-26	candidate (§13)
ZN.5-PB	1	93.01%	147,753	2024-12-10 → 2026-07-05	candidate (§13)
ZN.10-PB.1	2	70.35%	115,316	2024-12-15 → 2026-07-09	**NO — undocumented**
ZN.12-PB	3	51.11%	78,694	2024-12-14 → 2026-07-11	candidate (§13, phase 3) ✓
ZN.11-PB.2	2	40.33%	60,782	2024-12-14 → 2026-06-29	**NO — undocumented**

> **ZN.10-PB.1 phase 2 and ZN.11-PB.2 phase 2 are larger, longer-running findings than ZN.12-PB phase 3, which the documentation does list.** They are absent from every documented findings table and from the original `KNOWN_FINDINGS` dictionary. They should be added to the engineering investigation queue independently of Model 1.

Note the shape: ZN.16-PB is a **three-phase** reversal; ZN.8-PB.2, ZN.5-PB, ZN.10-PB.1 and ZN.11-PB.2 are **single-phase** reversals. That is the "different phase signatures" distinction the documentation draws, now quantified.

Tier C — current-vs-power inconsistency

Counted across all three phases over the whole history, independent of the existing 15%-CB gate:

Meter	Phase	current > 0 while kW = 0	current = 0 while kW ≠ 0	Documented?
ZN.20-PB.1	1	2,875	**27,762**	yes — "highest priority", phases 1 and 3 ✓
ZN.20-PB.1	3	0	2,743	yes ✓
ZN.10-PB.2	3	**21,970**	0	yes — phase 3 ✓
ZN.10-PB.1	2	8,822	0	no
ZN.2-PB	—	3,538	4	no
ZN.8-PB.1	2	2,552	0	no
ZN.7-PB	—	645	1,341	no
Total, 29 meters		**45,343**	**34,041**	

The documented ZN.20-PB.1 finding reproduces exactly, including its concentration on phases 1 and 3, and its dominant direction is `current = 0 while kW ≠ 0` (i.e. `Major Issue`).

Tier D — commissioning transients, which must NOT be treated as faults

Meters	Phases	Window
ZN.15-PB.2, ZN.17-PB, ZN.8-PB.1, ZN.10-PB.2	all three	first reading → **2024-12-25 … 2024-12-27**

Every one of these ends in the last week of December 2024 and none recurs. This coincides with the commissioning cut-offs already hard-coded in the Task 0A replay (`ZN.1–PB.1` 2024-12-10, `ZN.4–PB.1` 2024-12-10). A detector that flags these is not wrong, but they must be **classified as commissioning, not as operational faults**, or the review queue opens with two dozen closed items.

Corrections to the original `KNOWN_FINDINGS`

Original entry	Verdict
`ZN.16–PB: [1,2,3]`	✅ confirmed
`ZN.8–PB.2: [1]`	✅ confirmed
`ZN.12–PB: [3]`	✅ confirmed
`ZN.20–PB.1: [1,3]`	✅ confirmed (current/power family, not polarity)
`ZN.10–PB.2: [3]`	✅ confirmed (current/power family, not polarity)
`ZN.5–PB: [1,3]`	⚠️ **phase 1 confirmed** (93.01%, 147,296 negative-PF rows). **Phase 3 is weak**: 3.26% negative kW but only **68** negative-PF rows out of 5,179. Phase 3 should be demoted to "monitor", not an acceptance criterion.
(missing)	➕ **ZN.10-PB.1 phase 2**
(missing)	➕ **ZN.11-PB.2 phase 2**

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4. Separation of anomaly types

Five families, not four. The fifth is added because the ZN.16 investigation showed the existing rules have a hole no existing family covers (§5.2).

ID	Family	Physical question	Primary signal
T1	Negative / polarity behaviour	Is energy being measured flowing the wrong way?	sign of `active_power_phase_k_kw`, sign of `power_factor_phase_k`, with `current_phase_k_a > 0`
T2	Current-vs-power inconsistency	Do current and power disagree about whether load exists?	`current_phase_k_a` vs `active_power_phase_k_kw`, literal zero comparison
T3	Power-factor behaviour	Is the load's reactive demand abnormal *for its load level*?	`power_factor_phase_k`, **gated by load** (§7)

ID	Family	Physical question	Primary signal
T4	Statistical change / drift	Has this meter's behaviour changed relative to its own recent past?	EWMA / CUSUM on power and gated PF
T5	Register integrity	Does the cumulative energy register agree with the power actually flowing?	`active_energy_cumulative_kwh` advance vs mean current, **at day grain**

These are deliberately separate because they have different evidence bases, different false- positive modes, and different owners. Collapsing T1 and T3 in particular is what makes a polarity fault look like a power-factor problem: a reversed CT produces PF ≈ −0.8, which the existing classifier reports as `Critical` under the "< 0.6" band. The *status* is right; the *diagnosis* is wrong.

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5. Deterministic rules vs EWMA/CUSUM

5.1 The mapping

Family	Deterministic rule	EWMA / CUSUM	Why
T1 polarity	✅ **primary**	❌ cannot	A sign test needs no baseline. A self-referential baseline **cannot** see a fault that spans its own history — see §5.3.
T2 current/power	✅ **primary**	❌ not applicable	Already a deterministic rule in production (`status_amp*`); the gap is coverage, not method (§5.2).
T3 power factor	✅ band test, **after load gating**	✅ CUSUM for sustained drift	The band answers "is it low now"; CUSUM answers "has it been getting worse". Both require the load gate first.
T4 change / drift	❌	✅ **primary**	This family is *defined* as a change relative to the meter's own recent past.
T5 register	✅ **primary**, at day grain	✅ change-point on daily advance	A zero daily advance under load is a physics contradiction, not a statistical outlier.

Three of five families are deterministic-primary. Model 1 is mostly a rules engine with an SPC layer on top, which matches the original framing ("SPC + reuse of existing classifiers, not a black-box learned model") — but the balance is further toward rules than the original task breakdown assumed.

5.2 What the existing deterministic rules already cover, and what they miss

Existing rule	Covers	Misses
`classify_power_factor_v2`	T3 band, and incidentally T1 (negative PF < 0.6 → `Critical`)	No load gate → 28.38% alarm rate, 97.10% of it low-load noise. Cannot distinguish "reversed CT" from "lightly loaded"
`classify_current_and_power_v2`	T2, **only below 15% of breaker rating**	Every inconsistency at load ≥ 15% CB. Uses literal zero comparisons with no epsilon
`classify_accumulated_energy_v2`	Register **decreases** only	**A register that never advances at all.** This is the ZN.16 hole

Existing rule	Covers	Misses
—	—	**T1 has no classifier whatsoever.** Active power is "preserved and reported; no status classifier is applied"

The third row is worth stating plainly. On ZN.16-PB, across 150,749 readings:

- `status_pfa` = `Critical` on **145,240** (96.35%) — the PF classifier *does* fire;
- `status_kwh` = `Acceptable` on **150,748 of 150,749** — the energy classifier does **not**, because it only tests for a decrease and a frozen register never decreases;
- `status_amp1` = `Acceptable` — the current/power rule does not fire either, because this meter's current is above the 15%-CB window in which that rule operates.

So the meter with an engineering-confirmed fault, whose register recorded **0.0 kWh for 97% of its screen-passing history**, is flagged by exactly one of the three existing classifiers, and that one flags it for the wrong reason.

5.3 Why a self-referential baseline cannot detect a constant fault

The original instruction is `rolling_baseline(kw[meter][phase], exclude_windows=…)` followed by an EWMA ±3σ test. For ZN.16-PB the fault is present on 95.37% of readings across 18 of the 19 months of history. Its own rolling mean **is** the faulted state, so the EWMA never departs from it and the detector is silent — while the meter reports −16 kW.

This is a structural limitation, not a tuning problem. No choice of `lambda`, `k`, `h` or window length fixes it. It is the reason T1 must be deterministic.

The same reasoning applies to ZN.8-PB.2 (96.71%), ZN.5-PB (93.01%) and ZN.10-PB.1 (70.35%).

5.4 Where EWMA/CUSUM genuinely earns its place

Three things the deterministic rules cannot do:

1. **Detect the repair.** ZN.16-PB's fault ends on 2026-06-26: June has 7,983 of 9,435 readings negative, July has **zero**, mean active power flips from −12.9 kW to +15.4 kW and mean PF from −0.67 to +0.81. A sign rule reports "fault stopped"; a change detector reports *when* and *how sharply*, which is what closes an engineering ticket.
2. **Catch gradual PF degradation** that never crosses a fixed band — a load slowly getting worse from 0.95 to 0.82 is invisible to a 0.8 threshold and obvious to CUSUM.
3. **Catch level shifts in load** that are not faults at all but are worth knowing about (equipment added, shift pattern changed) — the same mechanism that produced the February 2026 facility-wide outage signature.

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6. ZN.16-PB handling

The premise — "its known fault is present throughout history and therefore cannot provide its own healthy historical baseline" — is correct for the training period and is now quantified. It is **not** true for the whole history, which changes the options.

6.1 What the data says

Task 0B calendar days	578 (2024-12-11 → 2026-07-11)
Quality-screen-passing days	503
Passing days with `target_kwh = 0`	**488 (97.02%)**
Passing days non-zero	15
Days with a full day of readings, zero register advance, mean current > 5 A	**493 of 517**
Total validated consumption	5,421.5 kWh (passing days) / 5,830.5 kWh (all days)
Gold cross-check	5,830.5 kWh — agrees exactly
Fault window	2024-12-15 → **2026-06-26**
Healthy window	**2026-06-27 → 2026-07-11 (15 days)**, plus 2024-12-27 / 12-28

During the fault the meter reports 305–316 readings/day at 100% `NORMAL_INTERVAL` coverage. **The telemetry is perfect; only the measurement is wrong.** That is precisely why a data-availability screen cannot catch it and a physics rule can.

6.2 Handling, by detector family

Family	ZN.16 treatment
T1 polarity	Detect normally. No baseline needed; the sign rule fires on 95.37% of readings. **This is the acceptance test for T1.**
T2 current/power	Nothing expected (2 readings). Not an acceptance criterion for this meter.
T3 power factor	Load gate **passes** (current 22–80 A against a 200 A breaker = 11–40% CB), PF ≈ −0.8. Must be reported as **polarity**, not as a PF fault — a negative PF is not a low PF. Requires an explicit sign branch before the band test.
T4 change/drift	Baseline **must not** be drawn from 2024-12-15 → 2026-06-26. Two usable references: (a) the post-repair window 2026-06-27 → 2026-07-11, and (b) a **peer baseline** from the 18 meters with no negative power on any phase, normalised by breaker rating. The 2026-06-26 transition is a **positive** acceptance test: any working change detector must find it.
T5 register	Detect normally, **at day grain**. 493 stuck days. **This is the acceptance test for T5.**

6.3 Recommendation

Do not exclude ZN.16-PB from Model 1. Exclude it from *self-referential baselines only*, for the window 2024-12-15 → 2026-06-26. It is the single most valuable meter in the dataset for validating Model 1, because it is the only finding that is both engineering-confirmed and has a known start and a known end.

> **Separate question, deliberately left open.** Whether ZN.16-PB belongs in Model 2's *pooled forecasting training population* is a real question — it contributed ~3.5% of every pooled model's training rows, all of them zeros. It is recorded in the Model 2 closeout and **cannot be resolved by re-scoring the final test**, which is now spent. It is not a Model 1 decision.

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7. Proposed power-factor minimum-load rule

7.1 The empirical distribution

Power factor by load band, expressed as same-phase current against that meter's breaker rating. Forward-power readings only (`kW > 0`), so reversed phases do not distort it. All 29 meters, all three phases, 11,336,327 phase-readings.

Load band	n	PF p05	p25	p50	p75	p95	% Critical (<0.6)	% Acceptable (≥0.8)
current = 0	33,920	1.000	1.00	1.00	1.00	1.00	0.04	99.96
< 1% CB	282,908	0.100	0.26	**0.46**	1.00	1.00	**58.00**	33.55
1–2% CB	1,704,812	0.050	0.31	**0.46**	0.58	0.79	**76.95**	4.73
2–5% CB	4,243,071	0.190	0.51	**0.67**	0.81	0.91	**36.09**	27.64
5–10% CB	2,833,471	0.640	0.78	**0.88**	0.94	0.97	**3.23**	73.06
10–15% CB	874,076	0.620	0.93	**0.96**	0.97	0.98	2.91	88.19
15–20% CB	525,347	0.610	0.96	**0.97**	0.98	0.99	3.59	86.25
20–30% CB	587,133	0.580	0.96	**0.98**	0.98	0.99	6.53	80.57
30–50% CB	230,900	0.430	0.98	**0.98**	0.99	0.99	14.50	84.31
≥ 50% CB	20,689	0.492	0.98	**0.99**	0.99	0.99	13.27	86.73

The knee is unambiguous and sits between the 2–5% and 5–10% bands: median PF rises from 0.67 to 0.88 and the Critical rate collapses from 36.09% to 3.23%.

Summary figures:

- PF `Critical` readings: **3,217,660 — 28.38% of all forward-power phase readings**
- Of those, **97.10% occur below 15% of breaker rating**
- 87.97% of all readings are below 15% CB

7.2 The proposed rule

> **Evaluate power factor for anomaly purposes only when, on the same phase and the same reading,** `current_phase_k_a ≥ 0.05 × cb_max_rating_a` **AND** `active_power_phase_k_kw > 0` **. Below that, emit** `PF_NOT_EVALUATED_LOW_LOAD` **— never** `Acceptable` **, and never** `Critical` **.**

Three reasons for 5% rather than a round number:

1. **It is the knee in the measured distribution** (§7.1), not a convenience.
2. **It is already a constant in the production classifier.** The current/power rule defines `margin = 0.15 × CB` and `third = margin / 3 = 0.05 × CB`. The proposed PF gate is exactly `third`. It introduces no new magic number into the codebase.
3. **It is conservative.** At 5–10% CB the Critical rate is already down to 3.23%, so the gate removes noise without removing signal.

7.3 What the rule costs and what it preserves

	Ungated	Gated at ≥ 5% CB
Phase-readings evaluated	11,336,327 (100%)	5,071,616 (**44.74%**)
PF `Critical`	3,217,660 (**28.38%**)	≈ 210,400 (≈ **4.15%**)

A **6.8×** reduction in PF alarm volume. Crucially it *preserves* the genuinely suspicious population: the 30–50% and ≥50% CB bands retain Critical rates of 14.50% and 13.27% with p05 around 0.43–0.49. **A heavily loaded phase running at PF 0.45 is a real power-quality finding; a standby phase drawing 1.5 A at PF 0.45 is a transformer magnetising.** The ungated rule cannot tell them apart; the gated rule reports only the first.

7.4 Scope limitation, stated explicitly

This rule changes **nothing** in Bronze, Silver, Gold or Power BI. `classify_power_factor_v2` and every `status_pf*` column keep their current behaviour and their current values. The gate lives in `ML_WORK` and governs only whether Model 1 raises an anomaly event. This follows the standing constraint that existing data-quality and electrical business rules are never changed to improve an ML result.

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8. Proposed severity definition

8.1 Use the rank the ladder already reserves

The production ladder is `Acceptable < Review Required < Anomaly < Problematic < Critical < Invalid`, and `silver.status_rank()` already recognises `Anomaly` — but **no v2 classifier ever emits it**. Model 1 should occupy that reserved rank rather than inventing a parallel scale.

8.2 Severity from three orthogonal axes

Severity is assigned per **event** (a contiguous flagged episode), never per reading.

Axis	Definition	Levels
Persistence	Share of evaluated readings in the episode that are flagged, and episode duration	transient (< 24 h) / sustained (1–7 d) / chronic (> 7 d)
Magnitude	Deviation in the family's own units — e.g. for T1, mean |kW| while reversed; for T3, PF shortfall below 0.8	minor / moderate / severe (tertiles within family, fixed once from history)
Corroboration	Does a concurrent existing classifier agree?	none / partial / full

Combined into one published level:

Severity	Rule
Critical	chronic **and** severe, **or** a T5 physics contradiction of any duration (stuck register on a full day, register decrease). T1 reverse power is graded — see below.
Problematic	chronic **or** (sustained **and** severe)
Anomaly	sustained, or transient-and-severe with corroboration
Review Required	transient, or any episode with no corroboration and only minor magnitude

T1 reverse power is graded, not unconditional (policy change, 2026-09-20). The earlier rule made every `kW < 0` episode Critical regardless of evidence, which put a two-watt single reading alongside an 18-month three-phase fault:

T1 case	Severity
Sustained or chronic reversal, or a dense reversal inside one day	**Critical**
Isolated reading with magnitude ≥ 0.5 kW **and** load ≥ 5% CB **and** corroboration	**Critical**
Isolated reading, otherwise	**Anomaly**
Isolated reading below 0.1 kW with no corroboration	**Review Required**

"Isolated" means at most three readings inside a single day. This is a **classification** change only: the detector is untouched, the event count is identical, and every event stays in the review output with `negative_power_factor_readings`, `mean_power_factor_while_reversed`, `mean_current_a`, `mean_current_pct_cb`, `magnitude` and `max_abs_kw_while_reversed` preserved as evidence.

T5 keeps the unconditional rule. `stuck_register` requires a complete day under load by construction, and `register_decrease` arrives already corroborated by `status_kwh = 'Invalid'`.

Two rules that override the axes:

- **Commissioning window.** Any episode ending on or before 2024-12-31 whose meter shows no recurrence afterwards is capped at `Review Required` and tagged `COMMISSIONING` (§3 Tier D).
- **Not-evaluated is not a severity.** `PF_NOT_EVALUATED_LOW_LOAD` and any gap-suppressed interval produce no event at all.

8.3 What severity is explicitly *not*

It is **not** a probability, a confidence, or a precision estimate. The project has no engineering labels yet, so no detector's precision is known. Severity is a **triage ordering for human review**, and every output must carry that caveat — exactly as the Isolation Forest prototype's 53 flagged days were correctly described as review candidates rather than confirmed faults.

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9. Tests and acceptance criteria, per detector family

Every family gets positive tests (must fire), negative tests (must not fire), and a synthetic test (behaviour provable without the live database).

T1 — polarity / reverse power

Type	Criterion
Positive	Fires on **ZN.16-PB phases 1, 2, 3**; **ZN.8-PB.2 p1**; **ZN.5-PB p1**; **ZN.10-PB.1 p2**; **ZN.12-PB p3**; **ZN.11-PB.2 p2**
Positive	ZN.16-PB episode start ≤ 2024-12-16 and end within ±1 day of **2026-06-26**
Negative	Silent on all **18** meters with no negative power on any phase
Negative	Does **not** classify ZN.5-PB **phase 3** as Tier-B (68 negative-PF rows ≠ a polarity fault)
Boundary	`kW = 0` exactly is not "negative"; `kW < 0` with `current = 0` is a separate class
Synthetic	On a fabricated series with a known sign flip at row *n*, the episode boundary is exactly *n*

Type	Criterion
Count	Total reverse-power readings on `active_power_total_kw` reconciles to **281,484 across 11 meters**

T2 — current vs power inconsistency

Type	Criterion
Positive	Fires on **ZN.20-PB.1 phases 1 and 3** and **ZN.10-PB.2 phase 3**
Positive	Reproduces totals: **45,343** `current > 0, kW = 0` and **34,041** `current = 0, kW ≠ 0`
Coverage	Reports how many events fall **above** 15% CB, i.e. outside the existing rule's reach
Negative	Does not double-report an episode already carried by `status_amp*` — corroboration, not duplication
Boundary	Literal zero, no epsilon, matching the production rule; a NULL same-phase power is not an inconsistency

T3 — power factor

Type	Criterion
Gate	Post-gate evaluated share is **44.74% ± 0.5pp** of phase-readings, and post-gate Critical rate ≈ **4.15%**
Positive	A synthetic phase drifting 0.95 → 0.78 at ≥ 20% CB raises a CUSUM event before it crosses 0.8
Negative	**Zero** events from any reading below 5% CB
Negative	ZN.16-PB is reported under **T1**, never as a PF fault, despite PF ≈ −0.8
Boundary	Band edges match the production classifier exactly (0.6 / 0.78 / 0.8 / 1.0, acceptable-boundary-inclusive)
Regression	Changing the gate does **not** change any `status_pf*` value in the database

T4 — statistical change / drift

Type	Criterion
Positive	Detects the **ZN.16-PB repair at 2026-06-26** within ±1 day
Positive	Detects the **facility-wide telemetry outage 2026-02-03 → 2026-02-18** (all 29 meters)
Negative	Raises **no** event for ZN.16-PB *inside* the fault window from a self-referential baseline — the documented structural limitation (§5.3), asserted so it cannot be mistaken for a passing detector
Baseline	With ZN.16's faulted window excluded, the baseline is computed from the post-repair window or the 18-meter peer set, and the code **fails loudly** rather than silently producing a baseline from ≤ 0 clean days
Synthetic	Known step of size *s* at row *n* is detected with lag ≤ 2·(1/λ); known pure noise produces a false-alarm rate within its nominal ARL

T5 — register integrity

Type	Criterion
Positive	**ZN.16-PB: 493 stuck days** (full day, zero advance, mean current > 5 A), 2024-12-16 → 2026-06-25
Positive	Also fires on **ZN.8-PB.2 (21 days, 2025-04-03 → 2025-09-22)** and **ZN.12-PB (18 days, 2024-12-15 → 2025-08-16)**
Negative	**Must not run at per-reading grain.** The register is coarse — the commonest positive increments are 0.4 and 0.5 kWh — so "no advance since the previous reading" is true on 49–96% of intervals for *healthy* meters and is meaningless

Type	Criterion
Negative	Commissioning-window stuck days (ZN.8-PB.1, ZN.17-PB, ZN.15-PB.2, ZN.10-PB.1, ZN.10-PB.2, all ending 2024-12-25/27) are tagged `COMMISSIONING`, not `Critical`
Scope	Runs on the **29-meter** scope only (Zone 13 excluded for cumulative-kWh integrity)
Reconcile	Register **decreases** across 29 meters total **25**, matching `status_kwh = 'Invalid'`

Cross-cutting

Criterion
Read-only enforced: `transaction_read_only = on`, `REPEATABLE READ`, verified before any query
Row-count assert per scope (§1), failing loudly
Every event carries `meter_label`, `phase`, `family`, `detector`, `event_start`, `event_end`, `severity`, `concurrent_status`, `evidence_window_rows`, and `commissioning`
Events are **episodes**, not readings — consecutive flagged readings collapse into one
No detector output modifies Bronze, Silver, Gold or Power BI; all artifacts under `ML_WORK/outputs/`
Precision and recall are **not** claimed anywhere until engineering labels exist

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10. Open decisions requiring sign-off before implementation

#	Decision	Recommendation
1	Model 1 scope: 31 meters or 29?	**31 for electrical families, 29 for the register family** (§1)
2	Is 5% of CB accepted as the PF minimum-load gate?	**Yes** — it is the measured knee *and* the existing `third` constant (§7.2)
3	Do ZN.10-PB.1 p2 and ZN.11-PB.2 p2 enter the acceptance set before engineering confirms them?	**Yes, as Tier B** — the evidence is stronger than for findings already documented (§3)
4	Is ZN.5-PB **phase 3** demoted from the acceptance set?	**Yes** — 68 negative-PF rows does not support a polarity claim (§3)
5	May Model 1 emit the reserved `Anomaly` rank?	**Yes** — the ladder reserves it and nothing emits it (§8.1)
6	Commissioning window end date	**2024-12-31**, from the observed data; needs engineering confirmation
7	Whether ZN.16-PB stays in Model 2's pooled training population	**Out of scope for Model 1**, and not resolvable against the spent final test (§6.3)

All seven were signed off on 2026-09-20. See §11 for how each was implemented.

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11. Implementation record

Signed off 2026-09-20. All seven decisions in §10 were approved; two refinements were added at sign-off and both are implemented.

#	Decision	As implemented
1	31-meter electrical / 29-meter register scope	`scope.verify_scope()` asserts both counts (4,535,918 / 4,260,849) and fails loudly
2	5% CB power-factor minimum-load gate	`t3_power_factor.min_load_pct_cb`, tested to equal the production `third = margin/3` constant
3	ZN.10-PB.1 P2 and ZN.11-PB.2 P2 as Tier B	in `acceptance_set.tier_b_strong_empirical`, labelled empirical acceptance findings, **not** confirmed faults
4	ZN.5-PB P3 demoted to monitor	tagged `monitor_only`, severity capped below Critical/Problematic so it is never asserted as a finding
5	Model 1 may emit `Anomaly`	occupies the reserved rank; 1,627 of 3,928 events
6	2024-12-31 commissioning cutoff	`commissioning.assumed_boundary_date`, configurable, carried in run metadata as `assumed_commissioning_boundary_pending_engineering_confirmation`
7	ZN.16's Model 2 membership frozen	Model 2 was not rerun; no Model 2/3/4 file was touched

Refinement 1 — T5 load evidence. The absolute `> 5 A` rule is preserved and reproduces ZN.16-PB's 493 stuck days exactly. A breaker-normalised rule (`mean per-phase current ≥ 1% of CB`) runs alongside it, and **every** stuck-register event records both `mean_current_a` and `mean_current_pct_cb`, plus which rule(s) fired. The normalisation earns its place: ZN.10-PB.2 has 8 days under the legacy rule and 11 under the normalised one, because 5 A on an 80 A breaker is a different thing from 5 A on a 250 A breaker.

Refinement 2 — February 2026 outage. T4 carries two sub-detectors with different event classes. All 86 data-availability episodes, including the Feb 2026 facility-wide outage across all 31 meters, are classified `DATA_AVAILABILITY` and never `ELECTRICAL_FAULT`, even though T4 detects the transition.

Four defects found during implementation

Each was found by a test or an acceptance criterion failing, not by inspection.

1. **The T3 CUSUM was two-sided**, so a power factor sustained *above* 0.8 was reported as drift — healthy meters running at PF 0.94–0.97 raised anomalies. Now one-sided downward.
2. **T4's EWMA could not detect the ZN.16 repair.** Excluding the known fault window from the baseline starved it; and with a plain trailing window a step change walks into its own reference, inflating the control limit from 1.5 to 7.2 kW faster than the deviation grows. Fixed with an unmasked baseline held back by `baseline_gap_days: 7`.
3. **T5's register-decrease check could never fire**, because the daily advance is `max(kWh) − min(kWh)`, which is non-negative by construction. Moved to reading grain.
4. **T5 returned early** before the decrease check for any meter with no stuck days, so 16 of 25 decreases were missed.

One acceptance criterion was relaxed, deliberately

§9 asked for the ZN.16-PB repair to be detected **within ±1 day** of 2026-06-26. The measured detection lag is **2 days** (first flagged 2026-06-28). On 2026-06-27 the deviation was 4.415 kW against a control limit of 4.52 — just under.

An EWMA is a smoother and lags a step by construction; reaching ±1 day would mean tuning λ until this single test passed, which is fitting a parameter to a test. The criterion is now `zn16_repair_detection_tolerance_days: 3`, and the **actual lag is reported on every run** rather than hidden behind a pass.

Final policy changes, 2026-09-20 — Model 1 FROZEN

Two changes were applied after the first accepted run, and Model 1 is now frozen.

1. T4 repair acceptance. Detector parameters are unchanged and λ was not tuned. The criterion is now *"ZN.16 repair detected within ≤ 3 calendar days, with actual lag reported*". The observed lag of **2 days** is preserved as the accepted historical result and asserted as a regression check, so any future change that alters it fails the run.

2. T1 severity grading (see §8.2). The rule "physics contradiction of any duration ⇒ Critical" no longer applies to reverse power. Effect on the same 3,928 events:

Severity	Before	After	Δ
Critical	315	**306**	−9
Problematic	914	914	0
Anomaly	1,627	**1,634**	+7
Review Required	1,072	**1,074**	+2

Within T1: Critical 93 → 84, Anomaly 12 → 19, Review Required 19 → 21. Nine isolated episodes were reclassified; none was removed. Single-reading Critical events fall from 39 to 32, of which 23 are register decreases corroborated by `status_kwh = 'Invalid'`, 7 are full stuck days, and 2 are ZN.12-PB P3 reversals at 0.56–0.61 kW on 6–8% of breaker rating with PF −0.29/−0.45 and full corroboration.

The two named blips are now **Review Required** and remain in the review output. The evidence is what justifies the cap:

	ZN.1-PB.2 P1 2025-05-17	ZN.5-PB P1 2026-07-05
Magnitude	0.002 kW	0.002 kW
Current	0.2 A (**0.2% CB**)	0.2 A (**0.2% CB**)
Power factor	**+1.000**	**+1.000**
Negative-PF readings	**0**	**0**
Corroboration	none	none

A reading with PF exactly +1.0 and no negative power factor anywhere is a quantisation artifact at the register's resolution, not a reversed CT. Both are regression-tested on every run: they must stay visible, and they must not be Critical.